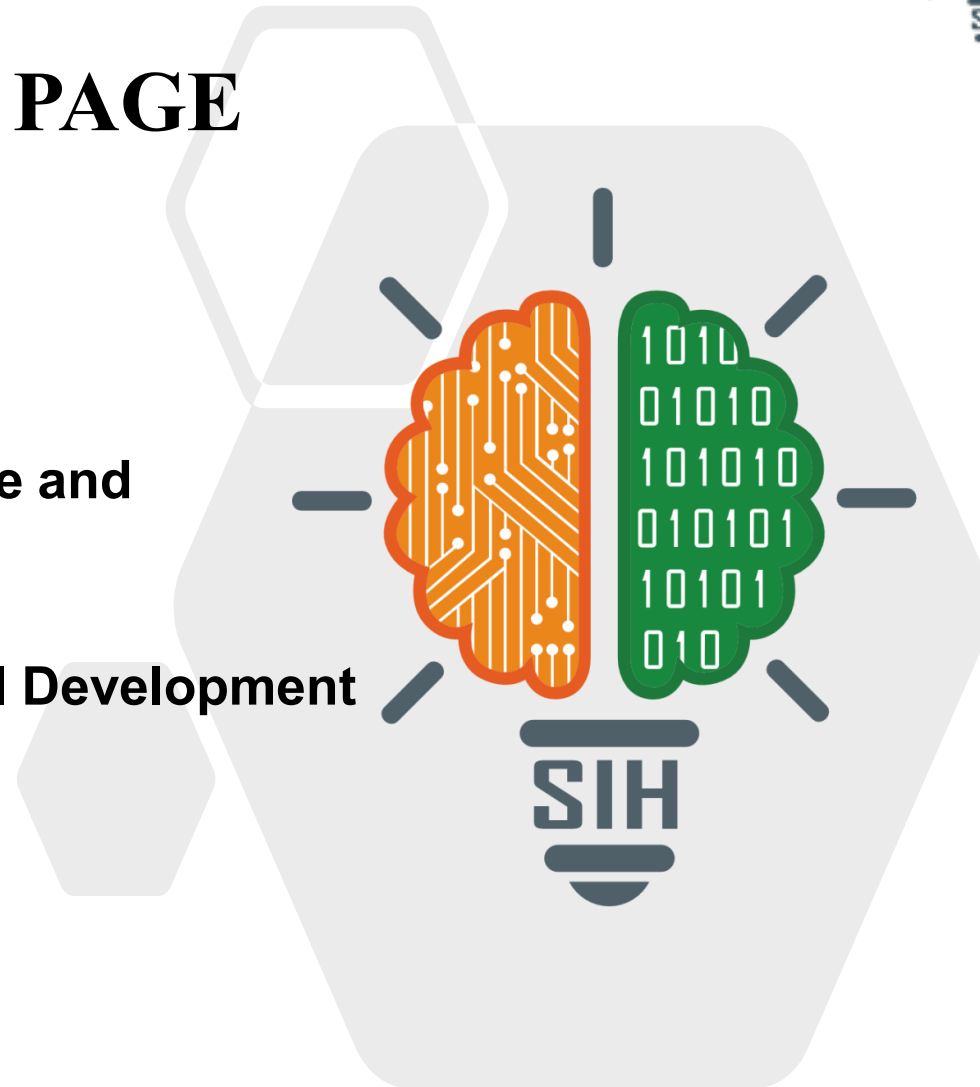


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID**—SIH25004
- **Problem Statement Title**-
Image based breed recognition for cattle and buffaloes of India
- **Theme**—Agriculture, FoodTech & Rural Development
- **PS Category**-Software
- **Team ID**- 64256
- **Team**-The Cattle Race



Idea Approach [1]

1. FLWs (Field Level Workers) capture the photo of cattle using their phones and upload it into the app.
2. The image is preprocessed .
3. The processed photo is passed sequentially through the species classification, body-segementation and breed detection models.
4. If the breed detection confidence is high , Top-3 predicted labels are displayed and stored in the database.
 - These curated, high confidence examples are used to retrain and improve the model.
5. If the breed detection confidence is low , the system requests textual inputs from FLWs like location , physical characteristics and uses these for better breed detection using Vision Transformers.
6. For a new unknown breed, the few shot learning routine is initiated .
 - 5-10 images are collected and model is finetuned on these images

Novelty/Uniqueness

- **Clean, high value database:** Only high quality, high confidence images are stored in the database
- **Cross-breed detection :** Detects animals that are genetic mixes of multiple breeds
- **Few shot detection of rare breeds:** Rapidly learns a new breed from 5-10 images to enable the continuous expansion of breed catalog

Addressing Key Feature Requirements

- **Accurate Breed Registration :** The AI model prevents mis-registration by only passing in high confidence image data into the database
- **Research & Policy Ready Data:** Verified records are stored in a knowledge graph for direct use in research and policy planning,
- **Robust Field Performance:** Image pipeline handles varied backgrounds, lighting, and animal poses to work reliably in real farm conditions.

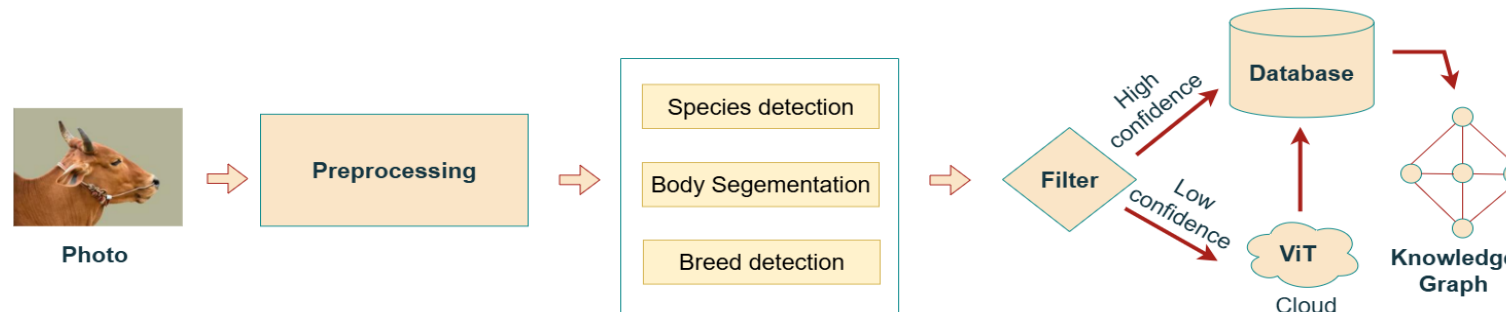
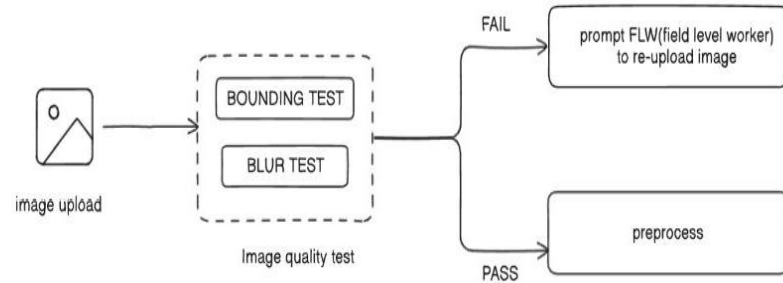


Image preprocessing



Breed Classification

1 : Breed classification using edge models

Detect species, segment body and predict top-K candidate breeds with confidence scores. Display GRAD-CAM and textual explanations using image captioning

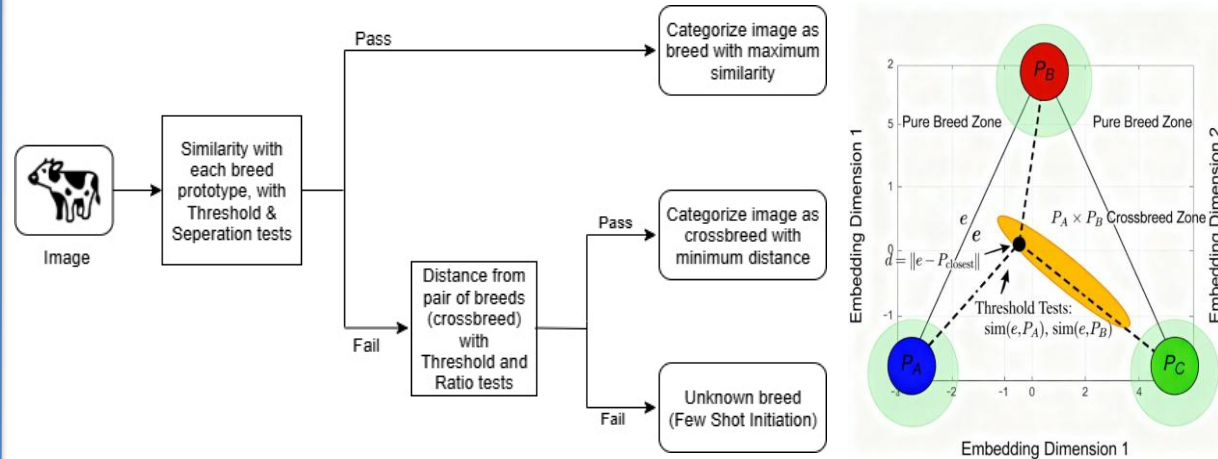
2: Conditional Cloud Refinement with Human-in-loop mechanism

When edge confidence is low, we interact with FLWs to extract the physical level characteristics of cattle and use them as an input to the vision transformer

3: Explainability

GRAD-CAM analyses various physical traits like horns, humps etc in an image. These highlights are then converted into short textual explanations using an image captioning model developed by us.

Cross Breed Detection [2]



Robust Image Performance

Trained on variety of lighting, backgrounds with breed and pose variety to simulate the real world



Diversity in Pose



Diversity in number



Diversity in Background



Diversity in Lighting



Occlusion

TECH-STACK

1 Efficient-NetB0
: edge model for breed classification

2 Vision Transformer :
Refines breed classification when edge confidence is low

3 Grad-CAM : Used to highlight visual traits in cattle for explainability

4 Image Captioning model : Generates short explanations using Grad-CAM regions

FEASIBILITY & VIABILITY

Economically feasible

Our system uses the well-developed techniques in edge deployment and lightweight AI models to create a economically feasible solution by keeping hardware and operation costs low.

Integrable with BPA

Our system is integrable with the BPA and therefore is scalable nationally

Operationally Viable

FLWs can use this system without a need for extensive technical training.

LIMITATION

Performance in extreme conditions

In cases like poor light , partial occlusion or extreme weather conditions the performance may degrade

Image Quality

Image Quality of the photos would depend on the technical device used by FLW in capturing of cattle

CHALLENGE

Differing field conditions

The system should accurately recognize across diverse set of poses , backgrounds and lighting conditions

Dependence on network availability

The FLWs may be covering an area with low network availability.

Detection of unknown/rare breeds

The system should handle breeds that do not match the breeds in the breed database

Research-ready data

The data should be recorded such that it can be used for research and policy purposed

SOLUTION

Overcomes the problem by integrating basic tests for image quality assessment which can then prompt FLW to reupload if the photo is extremely blurry or too dark to identify an object

The app is designed to work with basic network connection and will queue the images in low network areas, processing them as a batch when network improves

Few shot routine is initiated and new breed is added into the breed database with just 5-10 good images.

With addition of only high confidence images into the database , we build a knowledge graph which can be used by researchers to find new links and combinations

IMPACT AND BENEFITS

BENEFITS

Higher Data Accuracy

Visual explanations and quality checks reduce mis-registrations, which will lead to a more reliable database and enriched knowledge graph

Crossbreed Awareness

Avoids forcing single-breed labels on the image and calculates the cross breeds using probabilistic approach

Works in rural conditions

The system works even in low network areas with queued upload and batched processing

Few-shot expansion

The platform can keep up the pace with India's breed diversity without costly retraining of model using few-shot mechanism

Empowering with education

Textual explanations turn the app into a mobile classroom, allowing FLWs to build the skills while collecting the data

IMPACTS

Policy & Planning

Accurate, structured breed data supports better genetic-improvement programs, targeted disease control, and nutrition planning at a national scale.

Scientific Advancement

The knowledge graph enables large-scale studies on breed relationships, regional trends, and emerging crossbreeds.

National Standardization

Seamless BPA integration creates a consistent, country-wide standard for livestock data collection.

RESEARCH AND REFERENCES

REFERENCES

1. [A systematic review of machine learning techniques for cattle identification](#)
2. [RTDETR-Refa: a real-time detection method for multi-breed classification of cattle](#)
3. [Application of Convolutional Neural Networks in Animal Husbandry: A Review](#)

RESEARCH

1. [Detailed Idea Explanation](#)
2. [Crossbreed Detection](#)

YOUTUBE VIDEO LINK

